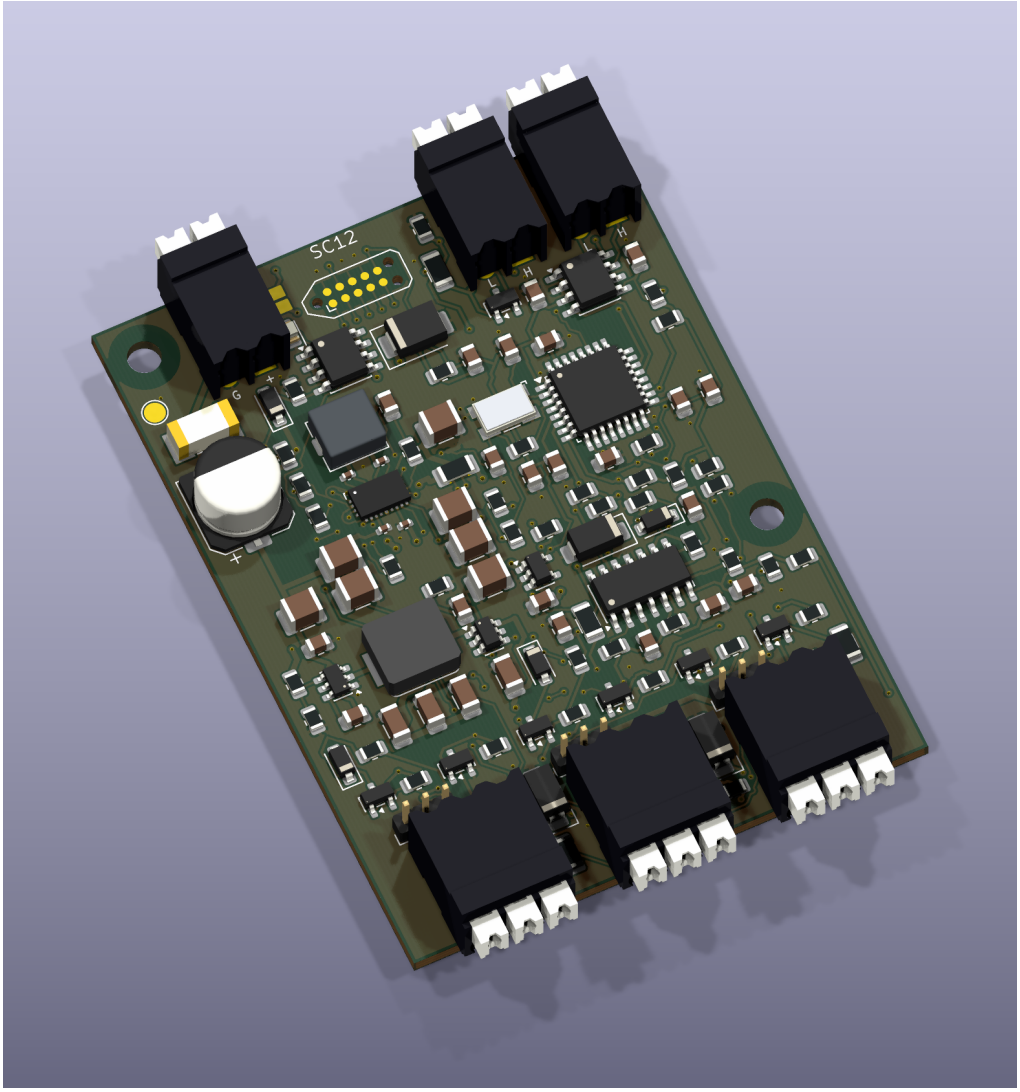
FRC
Sensor CAN

Revision E SC12 · Prototype engineering and assembly guide

Three industrial / logic sensor inputs
Regulated 12 V and 5 V sensor power



2026-09-08

Editable KiCad files, split factory/hand-solder BOMs, firmware, robot examples and verification records accompany this guide. No physical prototype has been tested. Electrical ratings are design targets pending the included qualification plan.

Contents

1 Product overview 3

2 Simplification 5

3 Wiring and programming contacts 7

4 Hand assembly 9

5 Firmware 10

6 CAN protocol 11

7 Robot integration 13

8 Purchasing and factory quote 14

9 Validation and qualification 16

10 Connector evidence 18

11 Tools and reproduction 19

12 Sources and notices 20

13 Cost reduction 21

14 Optimization progress and qualification 22

15 CAN setup and updates 24

16 Programming and unit fixture 27

17 Software input modes 28

1 Product overview

Source: README.md

Rev E adds independent **NPN, PNP, push-pull, floating-detection and keeper modes** through software. It removes the three input-bias solder selectors. The separate removable caps still choose 5 V or 12 V sensor power. How to configure and wire inputs.

This project serves a similar purpose to CANdi, with its own firmware, CAN protocol and robot drivers. It is **not CTRE Phoenix or GrappleHook compatible**. Earlier revisions remain preserved; use this revision's complete hardware/firmware package together.

Item	Rev E
Purpose	Three industrial digital sensor inputs over classic CAN
Board input	6-22 V DC at J1
Sensor power	Regulated 5 V or 12 V; 300 mA shared per rail
Signals	Positive 0-30 V; LOW ≤1.2 V / HIGH ≥2.4 V design targets
Bias	Per-channel software pull-up, pull-down, off, float detection or keeper
MCU / CAN	STM32F042K6T6 + TCAN3413; 1 Mbit/s
Firmware	App 0.6.0, bootloader 1.0.0; hardware 5, protocol 4
Setup/update	UID-based CAN discovery, ID/configuration, desktop GUI and recoverable CAN firmware upload; initial SWD installation
PCB	52 × 70 mm, four copper layers; In1 ground; In2 ground plus three bias-control routes
Assembly	All components on top; 111 factory SMT parts and nine user-soldered through-hole parts; three loose power caps

Unbuilt engineering prototype. The package includes CAD, firmware and engineering checks. Physical sensor compatibility, temperature, startup, EMC, surge, vibration and robot-bus behavior still require qualification. Floating detection cannot distinguish a broken wire from a normally inactive open-collector output.

Compared with D2, E adds six small MOSFETs and seven resistors, reuses the comparator and input-protection circuit, and extends the board by 6 mm to retain top-side assembly. The power stage, connector family and critical converter/crystal routing remain preserved. Circuit comparison.

Quantity	Parts purchase estimate per board
1	$56.42
5	$40.96
10	$35.14
25	$32.72

Includes supplier minimums, hand-fitted connectors/headers and loose caps; excludes PCB, assembly, attrition, freight and taxes. This feature adds about $0.45 per board at ten units. No delivered-price or one-month shipping commitment is established. Cost basis.

To commission a board, follow assembly, install the combined factory image through the custom SWD contacts, then discover it through CAN service. The desktop window offers mode and active-state choices for each input. CLI example from this directory:

```
python host/can_service.py --bridge localhost:5810 configure \
  --uid UID --device 17 --invert 0 --debounce 2 \
```

```
--modes npn,pnp,push-pull
```

Replace `UID` with the discovered serial. Sensor power is hardware-enabled; leave power caps off during initial bring-up and change them only with power disconnected. Input bias defaults off during reset/bootloader, then uses saved settings. Keep robot outputs inhibited during maintenance.

Deliverable	Link
Native hardware	PCB, schematic, schematic PDF
Factory upload	Gerbers, SMT BOM, JLCPCB BOM, CPL, complete packet
Hand-fitted parts	BOM, caps
Wiring and software	Wiring, input modes, firmware, drivers, protocol
Mechanical	STEP, drawing, connector details
Review	Interactive BOM, checks, handbook
Source/review archive	ZIP, manifest

Repository revision comparison · Tools · Sources and licenses

2 Simplification

Source: docs/SIMPLIFICATION.md

Rev E extends D2 from 52 × 64 to 52 × 70 mm (9.4% more board area) to add electronic bias while preserving top-side assembly. It has 120 fitted parts and three loose power caps. The existing regulated 12 V/5 V supplies and comparator input path remain.

What saves space and parts

• A four-switch synchronous TPS552872-Q1 integrates the buck-boost switches and output-current-limit control. It replaces the raw-battery rail and its separate TPS1H200A switch, current-set resistor and three capacitors.

• Its driver uses the existing internal 5 V buck rail, avoiding the high-input-voltage loss of its internal linear bias regulator. The 5 V design reserves 100 mA for this supply. Startup and brownout sequencing must be checked on the prototype.

• One 0.1 ohm 1206 resistor sets the 12 V current limit. Separate PCB Kelvin taps carry the feedback signals; they are copper features, not extra components. This resistor dissipates about 9 mW at 300 mA.

• Both internal copper layers are ground. Signal and power routing stay on the outer layers. The return planes and local ground vias help a compact switching layout; this is not a measured EMI claim.

• MCU internal pullups read the two open-drain health outputs without two external resistors. The converter's unused current-limit and cable-compensation status/functions are omitted.

• 10k compensation resistance reuses an existing BOM line. Firmware inversion/debounce, fixed input thresholds and one status LED retain Rev C's simpler interface.

Why the remaining capacitors matter

The converter has two bootstrap capacitors, two high-frequency supply bypasses, input/output energy-storage banks, a driver-bias capacitor and compensation capacitor. They serve different current paths. Moving or merging a bootstrap/bypass capacitor into a distant bulk capacitor is not equivalent.

The six added 10 µF 50 V X7R 1210 capacitors share one MPN. Samsung's typical bias curves plus explicit tolerance, temperature and aging allowances give approximately 7.10 µF for the two-cap input bank at 22 V, 15.32 µF for the three-cap output bank at 12.22 V, and 6.20 µF for driver bypass at 5.22 V. TI calls for at least 4.7 µF effective input, 10 µF output and 4.7 µF bias capacitance. These are engineering budgets from typical curves, not manufacturer-guaranteed minima. Substitution needs the same review.

The existing three 22 µF capacitors on the 5 V buck retain its effective-capacitance budget. Input TVS/filter/feedback parts handle industrial wiring and noise. Each MCU/CAN/comparator bypass remains at its local supply pins. Numerical checks record the assumptions.

Assembly compromise

There are 39 resistors, 37 capacitors and 31 other fitted parts. Only four capacitors are 0402; other small passives are largely 0805, with 1206/1210 parts where power or effective capacitance requires them. The converter is a 5 × 3 mm HotRod QFN with five long electrical power pads, not a generic exposed-pad QFN. Its footprint follows TI drawing RYQ0021B. A laser stencil, controlled reflow and joint inspection are required for repeatable assembly. This is less forgiving than Rev C's mostly leaded packages.

No per-port electronic power switch, output driver, encoder interface, ADC voltage/current telemetry, wireless radio or USB connector is added. All three ports share each rail's load budget. There are three removable 2 mm power headers, six software-controlled bias MOSFETs and one optional CAN termination bridge. The user hand solders all nine through-hole parts; the factory places 111 SMT parts. Procurement changes and cost basis.

Efficiency and stability scope

Nominal switching frequency is 781.25 kHz; automatic light-load mode is enabled. The shielded 3.3 µH inductor is sized for converter peak-current protection, not just the 300 mA load rating. The low-value shunt and external 5 V gate bias reduce avoidable loss. No measured efficiency percentage is claimed.

The boost continuous-conduction analytical model screens 192 corners using TI equations 19-26, capacitor derating, inductance tolerance, a ±20% transconductance allowance and an assumed full-cycle delay. It is not a switching simulation. PFM, current limiting, line steps, load steps, startup and the buck/boost transition still require measurement. See loop report and plot.

At these modest sensor loads, operation can enter discontinuous/light-load mode; the 192-corner CCM model is a screening model rather than a prediction of the normal light-load waveform. The 3.3 µH choice keeps the inductor small with adequate saturation margin, while automatic light-load operation avoids unnecessary circulating current. Measure ripple and input power with the actual sensors.

3 Wiring and programming contacts

Source: docs/WIRING.md

Use the included hardware-5/protocol-v4 firmware and tools; Rev D/D2 firmware must not be used on Rev E. Board coordinates below are viewed from the component side, origin at the top-left PCB corner. Bottom silkscreen is read after turning the board over. Do not infer pin numbering from left/right after flipping.

Connector	Pin 1	Pin 2	Pin 3
J1 power	Positive input, 6-22 V	GND	-
J2, J3 CAN	CAN H	CAN L	-
J4 input 1	Selected sensor power	GND	Signal
J5 input 2	Selected sensor power	GND	Signal
J6 input 3	Selected sensor power	GND	Signal

Each terminal pole has two solder legs; they are one electrical contact. J2/J3 are connected in parallel for bus continuation, with no onboard CAN ground terminal. Common ground comes through robot power. Use twisted H/L wiring; keep branches short. Strip 8 mm using the conductor range specified by Weidmüller for the exact terminal. Press the white release button for removal and for conductor types requiring it. Provide cable strain relief independently of the terminal.

Feed J1 from appropriately fused robot distribution. F1 is secondary board protection. Do not backfeed J4-J6 power pins, parallel the two sensor rails, or apply external power to the programming rail. The board is not galvanically isolated.

Sensor power and software bias

The factory leaves JP101/JP201/JP301 unpopulated for hand soldering. Fit the specified 2 mm headers, then leave caps off for initial bring-up. With power disconnected, fit ONE cap across 1-2 for 5 V or 2-3 for 12 V. No cap disconnects port power. Connecting all three power pins shorts the rails.

The headers are centered at X=7, 23 and 39 mm, Y=57.5 mm. Pins 1-2-3 run left to right from the component side; pin 1 is square. There are no physical input-bias selectors. Set each channel to NPN, PNP, push-pull, float detection or keeper in software; see input modes.

Use Harwin M22-2010305 headers and Sullins SPN02SVEN-RC closed-top caps (2 mm pitch; not 2.54 mm). Caps are friction-retained, not mechanically latched. After voltage/functional verification, support retention with a small removable bead of electronics-grade neutral-cure silicone between the cap exterior and header body, clear of the contacts. This is a proposed retention method; qualify it for your robot vibration/impact environment and enclosure. Allow full cure before use and inspect seating during service. Do not claim these are vibration-qualified merely because there is no slider. Bridge JP2 only at a CAN bus end that needs a 120 Ω termination. Leave it open at intermediate nodes; normally there are two terminations on the entire bus.

Examples: select `npn` for a sinking output (default active LOW), `pnp` for a sourcing output (active HIGH), or `push-pull` for an actively driven logic signal. NPN pull-up current is at most approximately 3.5 mA. Both NPN and PNP compatibility budgets allow 100 µA OFF leakage. No onboard 3.3 V sensor-power output is provided; the signal pull-up is not a sensor supply.

Design targets: 0-30 V positive input, LOW ≤1.2 V / HIGH ≥2.4 V, common-ground offset budget ±0.1 V. Do not use as an analog input, quadrature encoder, IO-Link port or continuous negative-voltage input. Sampling is 1 ms with 1-100 ms debounce; short pulses can be missed. Sensor power budgets are shared, 300 mA per rail. Rail-health indications are available; they are not calibrated voltage/current measurements. A fault on either rail makes the health-checked driver input unavailable. See protocol.

Representative industrial sensors

Regulated 12 V is a suitable nominal supply for the following families, subject to exact model and cable checks. Connect the selected output only; insulate unused complementary output wires separately. Brown/blue/black is common supply/ground/output coloring, but follow the specific datasheet rather than assuming colors.

Family	Published limits	Rev E setup / limitation
Omron E3Z three-wire NPN/PNP	12-24 V ±10%; typically 30 mA maximum, through-beam pair 35 mA; residual ≤1 V below 10 mA	Power 2-3 for 12 V. NPN or PNP software mode. Residual voltage fits the LOW target at this board's small load
Banner QS18 standard opposed/diffuse/retroreflective variants	10-30 V; <25 mA excluding output load; standard outputs <1 V at 10 mA	12 V supply is in range. Select NPN or PNP bias to match model; use one output. Account for its 100 ms power-up delay
Banner QS18 DVS/DL/FF variants	NPN residual specified up to 1.5 V at 10 mA	This exceeds the board's design LOW target. Prefer a compatible PNP variant or qualify actual NPN residual at the ~3.5 mA pullup load; do not assume all QS18 NPN models work

Sources: Omron E3Z specifications, Banner QS18 manual. PNP OFF leakage must also meet the input pulldown budget. Observe manufacturer cable/fusing requirements. Two-wire current-loop, AC, analog, IO-Link and safety/OSSD sensors are not supported as interchangeable three-wire inputs.

12 V is specified at the board, not guaranteed at the remote cable end under every transient. The analytical DC divider/reference budget is approximately 11.80-12.22 V before dynamic/distribution allowances; check ≥10.8 V at an E3Z during startup and worst battery/load conditions. Three 30 mA sensors are comfortably within the 300 mA shared design target, but include emitters, indicator loads, cable losses and inrush. The board's 100 ms rail qualification is not proof that every sensor has completed its own startup.

Programming contacts

J8 uses the **TC2050-IDC-NL mechanical pattern and a custom electrical mapping**, centered at (24, 5.75) mm. Use a mapped adapter, not an assumed ARM/STDC ribbon pinout. Ten 0.7874 mm contact pads are on 1.27 mm pitch, with three mechanical guide holes. In the component-side view, the upper row runs **10, 9, 8, 7, 6** left to right at Y=5.115 mm; the lower row runs **1, 2, 3, 4, 5** at Y=6.385 mm. Pin 1 is at X=21.46 mm, Y=6.385 mm. Use this U-shaped numbering, not an odd/even ribbon numbering assumption.

J8 pin	Signal	Connection
1	NRST	Probe reset, optional
2	MCU UART RX	3.3 V UART adapter TX
3	MCU UART TX	UART adapter RX
4	SWCLK	SWD clock
5	SWDIO	SWD data
6	+3V3	Probe VTref sense **only**, board output
7	GND	Probe ground
8	GND	UART ground
9	BOOT0	Normally leave open; onboard pulldown
10	GND	Ground

Power the board through J1 during programming. Leave adapter/probe power disconnected. BOOT0 is normally low. SWD programming and custom pogo-adapter fit require bench verification. The intended clip-free probe is held perpendicular by a fixture; the no-legs pattern does not retain itself.

4 Hand assembly

Source: docs/ASSEMBLY.md

Factory: fit only the 111 SMT references in Factory_BOM.csv using CPL_JLCPCB.csv. User: buy the nine through-hole parts and three closed-top caps.

Use the exact BOM, assembly drawing, interactive BOM and placement files. Reference labels are in these drawings; dense placement does not leave room to print every reference legibly on the actual top silkscreen.

1. Order the 52 ×70 mm, four-layer, 1.6 mm board with 1 oz finished copper target. Use the exported Gerbers and separate plated/nonplated drill files. Have the fabricator review its actual stackup, annular rings, mask registration and minimum features. Gold finish is useful at the programming contacts; keep solder and paste off them.

2. Check bare-board continuity and orientation. Apply paste to SMT pads only. A 0.10 mm stencil is a starting point; the assembler must review release and the exact QFN apertures. Through-hole header pads, solder selectors, TC2050 contacts and the test point deliberately have no paste.

3. Place 0402/0805/1206/1210 passives, diodes and ICs. Verify C1 polarity, TVS/diode cathodes, pin 1 on each IC and the exact latch-off TPS2553DBVR-1 suffix. U8 TPS552872-Q1 has five long electrical power lands and fine-pitch perimeter pads; it is not a generic single-exposed-pad package. Use its supplied paste pattern, a controlled reflow profile and suitable joint inspection. There are no components on the bottom.

4. Inspect joints and measure for supply shorts. Hand solder the six horizontal Weidmüller terminals after SMT reflow as the user assembly step. Then fit JP101/JP201/JP301: 2 mm headers on the top side, straight and fully seated, with pin 1 at the square pad. Use a removable cap temporarily as an alignment aid with the board unpowered; remove it afterward. Fill both plated holes per terminal pole. Support the board without loading fresh SMT joints. The terminal pins extend below the board.

5. Leave all selectors open for initial bring-up. Supply J1 from a current-limited bench supply, starting at 12 V and a modest current limit. Confirm input polarity, protected input, approximately 5.07 V internal, 3.3 V logic and 12.01 V regulated rails. Sensor headers remain unpowered while power selectors are open.

6. Program the combined factory image through the custom TC2050 mapping, set/verify boot protection, power cycle and configure a unique CAN ID using CAN service. Follow the fixture procedure. Save each input mode in software and verify readback. Then power off, choose each port's supply and termination settings, inspect one-cap-per-header placement, and check that 5 V and 12 V are not shorted.

7. Perform the qualification checks before connecting a robot harness. Begin with a known load and known logic source, then the actual sensor/cable.

Use M3 hardware at (3.5,4) and (48.5,42.5) mm. Holes are 3.2 mm NPTH. The copper keepout has at least 3.30 mm radius; use hardware whose contact area stays within a 6.4 mm diameter, with insulating washers/spacers as appropriate. Account for component bodies, connector overhang and protruding pins. Do not clamp the PCB between conductive surfaces. The mechanical drawing gives nominal geometry, not a fitted enclosure.

The compact board still benefits from magnification and a fine iron. Rev E restores removable voltage shunts and retains large passives where possible. See the retention and vibration qualification notes in WIRING.md. The buck-boost QFN and four 0402 capacitors make stencil/reflow work more demanding than Rev C; it is not a through-hole beginner kit.

5 Firmware

Source: firmware/README.md

Bare-metal STM32F042K6T6 firmware for three digital inputs. Build allocation is recorded by the current linker map/build report; stack depth has not been measured. The final two 1 KiB flash pages hold redundant 32-byte versioned configuration with CRC and commit-last writes.

Rev E uses **hardware identity 5** and **protocol 4**. It has a separate image tag and configuration magic so D2 images/settings are rejected. Use the matching drivers and software input modes. The separate power caps select sensor voltage.

CAN service is implemented, physically unqualified: full-UID discovery, identify LED, persistent ID/inversion/debounce and verified uploads. See CAN setup/update, robot bridge and qualification plan.

From `rev_e/`:

```
source tools/env.sh
make -C firmware
bash scripts/test_firmware.sh
```

Outputs: `build/factory.hex` for initial SWD installation; `build/application.sc12` for CAN updates; separate bootloader/application ELF, BIN and MAP files. `sensor_can.hex`/ELF are application-only at 0x08002000. Flash: 8 KiB bootloader, 21 KiB app, 1 KiB manifest, 2 KiB configuration. Both linkers reserve the first 256 SRAM bytes and a 1 KiB minimum stack. The MCU controls six bias gates; no ADC sensing, sensor rail GPIO control or per-input LED logic is provided. The 8 MHz external crystal and PLL generate 48 MHz; bxCAN uses 16 time quanta with an 87.5% sample point at 1 Mbit/s. Auto bus-off recovery is enabled; automatic transmit retry is disabled.

Power through J1 from a bench supply. Use the custom TC2050 wiring: VTref sense→6, SWDIO→5, SWCLK→4, ground→7, optional reset→1. Leave probe power disconnected.

```
openocd -f interface/stlink.cfg -f target/stm32f0x.cfg \
  -c 'adapter speed 1000' \
  -f tools/program_factory.tcl
```

This command has not been exercised on physical Rev E hardware. Verify MCU identity, supply and flash readback. Normal application programming should preserve the two configuration pages; full-chip erase removes them. Version-1/2 configuration is rejected.

UART: adapter TX→J8.2, RX→J8.3, ground→J8.8; 3.3 V, 115200 baud, 8N1; adapter power disconnected.

```
python host/sensor_can.py info --port /dev/ttyUSB0
python host/sensor_can.py configure --port /dev/ttyUSB0 \
  --device 17 --invert 1 --debounce 2
```

Device IDs 0-63 must be unique in this Team Use address space. Invert is a 3-bit mask; debounce is 1-100 ms. UART `SET id invert debounce_ms` and UID-targeted CAN service save persist and reset. Legacy CAN runtime configuration remains volatile. Unconfigured boards accept discovery/service but suppress normal telemetry. Power-cycle after initial SWD programming and verify boot protection before CAN updates. **Hardware sensor power is enabled independently of firmware**, including during reset or programming. On Rev E, leave the caps off JP101/JP201/JP301 to disconnect port power during initial bring-up; configure them only with input power disconnected. Input bias is electronic; CAN termination remains a solder setting. See assembly.

The independent watchdog is serviced only after the full 1 ms foreground cycle. A >20 ms foreground delay, clock failure, hard fault or PVD low-voltage detection puts CAN into standby; the watchdog provides reset recovery. The single LED pulses approximately 100 ms per second when configured and healthy, or 100 ms per 200 ms for configuration/clock/CAN errors. Rail faults also trigger the fast fault indication. PA3 reads 12 V PG and PA4 reads 5 V FAULT_N, both with internal pullups. A fault clears qualification immediately; recovery needs 100 ms continuously good indications. Counts freeze and debounce is re-baselined during this interval.

Broadcast Disable ID 0 stops telemetry and aborts pending transmissions. Local sampling and hardware sensor power continue. An explicit `RESUME` UART command or CAN resume command re-enables telemetry; reads never do this automatically. See protocol v4. Host tests cover firmware logic and mocked peripheral boundaries, not actual oscillator startup, electrical timing or STM32 peripheral behavior.

6 CAN protocol

Source: docs/CAN_PROTOCOL.md

1 Mbit/s classic CAN, extended 29-bit identifiers, 8-byte telemetry. This is an original Team Use protocol, not CTRE Phoenix. Device type=10 (Miscellaneous), manufacturer=8 (Team Use), unique device ID=0-63. Address = `(10 << 24) | (8 << 16) | (API << 6) | device`.

The APIs differ from A/B/B-2; Rev C shares these API numbers but uses protocol v2, which Rev E drivers reject. The version and capability checks prevent drivers from interpreting the wrong revision's payload. Provision each board by full-UID CAN service or 3.3 V UART. An unconfigured board suppresses telemetry but answers service requests; its sensor power hardware is still enabled.

All multibyte fields are little endian. Masks use bits 0-2 for inputs 1-3. Unsupported versions, reserved bits, wrong lengths or stale frames must be reported as unavailable.

API	Period	Eight-byte payload
0x120 status	10 ms	version=4, sequence, debounced input mask, raw mask, channels=3, capabilities=0x7F, flags, reset flags
0x121 counts 1/2	100 ms	two uint32 rising-edge totals
0x122 count 3	100 ms	uint32 total, uint32 reserved=0
0x130 diagnostics	100 ms	uint32 transmit drops, TEC byte, REC byte, LEC byte, CAN flags byte
0x132 input state	Same as status	version=4, valid mask, floating mask, classified HIGH mask, uint32 input configuration
0x131 identity	1000 ms	4, firmware major=0, minor=6, patch=0, hardware=5, device ID, debounce ms, inversion mask

Status flags: configured=1, clock healthy=2, CAN initialized=4, bus-off=8, qualified 12 V power-good=16, qualified 5 V no-fault=32, all input classifications settled=64. Reset flags: power-on=1, pin reset=2, watchdog=4, software=8. Capabilities bits 0-4 indicate edge counts, runtime configuration, UART persistence, standalone hardware power and broadcast disable. Bit 5 indicates rail-health reporting capability; bit 6 indicates software bias/classification. Capabilities are static; flags bits 4/5 carry live qualified indications. The raw mask is the classified HIGH mask, excluding float probe transitions; check the valid mask in 0x132 before interpreting it. The debounced mask applies the active-state condition, inversion and debounce. Sequence is an unsigned byte and wraps. Counters wrap at 2^32 and clear on reset; runtime configuration preserves totals.

Diagnostic LEC is the STM32 three-bit last-error code (0-7); CAN flags are bus-off=1 and error-passive=2. There is no calibrated sensor voltage/current telemetry. The 12 V PG signal has the converter's typical threshold and tolerances; 5 V no-fault is an overcurrent-switch indication, not proof of voltage. Raw masks remain diagnostic data during faults.

A bad rail indication clears its flag at the next 1 ms sample; restoration needs 100 consecutive good samples. During faults and recovery, firmware re-baselines input debounce and freezes edge totals, preventing supply collapse/recovery from adding counts. All ports require both rail indications because sensor-power cap positions are not readable. Java/C++ input helpers also require fresh telemetry, configuration, clock health and CAN health and all classifications settled. They cannot detect every failed/disconnected sensor or open cable.

Command API	Payload	Effect
0x220 runtime input config	`53 43 04 invert debounce_lo debounce_hi 00 00`	invert 0-7; debounce 1-100 ms; volatile, does not change ID
0x222 runtime bias	`53 43 04 00` + uint32 configuration	Volatile, re-baselines input classification; read back 0x132
0x221 resume telemetry	`53 43 04 00 00 00 00 00`	Explicitly resume after broadcast disable

Extended arbitration ID **0** is the FRC broadcast disable. Firmware stops telemetry and aborts pending transmit mailboxes immediately on processing it. Sampling and standalone sensor power continue; there are no actuator outputs. Resume requires the explicit command above, UART `RESUME`, or reset. Drivers never resume from a read loop. Integrate explicit resume with a reviewed robot lifecycle policy; this implementation still requires robot-bus testing and does not establish competition approval. See WPILib CAN addressing.

Read timeouts: status 100 ms, counters/diagnostics 350 ms, identity 1500 ms. Counters 1/2 and 3 are separate snapshots. Sampling is 1 kHz; debounce requires a continuously stable logical state for the selected interval. This is not a high-speed pulse counter or encoder.

UART: 115200 baud, 8N1, 3.3 V logic. Commands `INFO`, `HELP`, `SET id invert debounce_ms`, `RESUME`. `SET` saves a version-4 redundant CRC-protected record and resets. The old four-argument command and earlier flash records are rejected; provision Rev E explicitly. No power-control command exists.

Service extension in firmware 0.6.0

Rev E changes normal telemetry to protocol 4; use only the matching drivers. APIs 0x300/0x301 at address 0 are reserved for CAN service. Opening maintenance pauses telemetry/counting on participating SC12 nodes; treat samples as unavailable immediately in robot code. Status period can be 10/20/50 ms until reset.

Per-input configuration word

Bits 0-2, 3-5 and 6-8 select channels 1-3 mode: 0=OFF, 1=PULL_HIGH, 2=PULL_LOW, 3=FLOAT_DETECT, 4=BUS_KEEPER. Bits 9-11, 12-14 and 15-17 select each active condition: 0=HIGH, 1=LOW, 2=DRIVEN, 3=FLOATING, 4=NOT_HIGH, 5=NOT_LOW. Bits 18-31 must be zero; other enum values are invalid. Default word zero means push-pull/HIGH on all channels.

Input state 0x132: a cleared valid bit means UNKNOWN regardless of the retained high/floating bit. For a valid channel, floating takes its own state, high means HIGH, otherwise LOW. High/floating masks never overlap. Validity/classification precedes active-state mapping, inversion and debounce. See mode timing and electrical limits.

At the default 10 ms period, telemetry is 231 frames/s; 20/50 ms give 131/71 frames/s. Status and input state are separate snapshots. Read timeouts are 100 ms for both; 350 ms for counts/diagnostics and 1500 ms for identity. Runtime configuration does not reset counters or persist to flash. Measurements settle after a mode change before further edges are counted.

7 Robot integration

Source: examples/README.md

Java 17 and C++ examples target WPILib 2026.2.1. Rev E uses protocol **4**, hardware **5** and firmware **0.6.0**. Use the matching `SensorCan12` and `SensorCan12Codec` files; older revisions are rejected. Provision a unique device ID through CAN service or UART.

Choose each sensor's 5 V / 12 V supply using the physical jumper before operation; the drivers do not select port voltage. See the Rev E setup guide and wiring.

CAN discovery, persistent settings and firmware updates are provided by the new service tools. The robot bridge has separately compiled 2026 and 2027 alpha-7 adapters. Sensor driver classes below remain pinned to 2026.2.1; the 2027 bridge does not port the normal drivers. Physical end-to-end CAN testing is pending.

```
private final SensorCan12 sensorBoard = new SensorCan12(17);
// In periodic(): absent means unavailable or a reported fault.
var detected = sensorBoard.input(1);
if (detected.isPresent()) {
  boolean sensorActive = detected.get();
  // Use a reviewed subsystem policy for this observation.
}
```

C++: `sensorcan12::SensorCan12 board{17};` then `board.ReadInput(1)` returns `std::optional<bool>`. Construct the driver once; an empty result must not silently become a valid “no object” observation. The helper requires fresh status, configured identity state, healthy clock/CAN and both qualified power indications. Raw `status()` / `ReadStatus()` remains available to diagnose faults. Counts are frozen during power faults and wrap at 2^32; compare/reset policy in the robot application.

Reads do not transmit or resume telemetry. `configureRuntime` / `ConfigureRuntime` sends volatile input inversion/debounce; it does not set persistent ID. `resumeTelemetry` / `ResumeTelemetry` is an explicit action after investigating broadcast disable, not a periodic automatic retry. Status timeout is 100 ms; count/diagnostic timeout 350 ms; identity timeout 1500 ms.

The Python host tool supports UART `info`, `configure`, `resume` and passive CAN `monitor`. UART uses 3.3 V levels at 115200 8N1; disconnect adapter power. CAN monitor requires a suitable external CAN interface; a UART adapter is not a CAN interface. Run the following from `rev_e/`:

```
source tools/env.sh
python host/sensor_can.py configure --port /dev/ttyUSB0 --device 17 --invert 1 --debounce 2
python host/sensor_can.py monitor --interface socketcan --channel can0 --device 17
bash scripts/test_drivers.sh
```

Tests cover codec boundaries, all reserved status bits, stale/malformed responses, fault rejection, unsigned counts and explicit-only writes. The C++ example compiles against the actual pinned WPILib headers; Java compiles against pinned WPILib jars. Hardware bus behavior and competition acceptance remain open qualification steps. See protocol, wiring, and validation.

Independent input modes

Use the desktop service tool to persist per-input settings. Runtime Java example (volatile; verify `inputState()` readback):

```
sensor.configureInputsRuntime(
    new SensorCan12Codec.InputMode[] {
        SensorCan12Codec.InputMode.PULL_HIGH,
        SensorCan12Codec.InputMode.PULL_LOW,
        SensorCan12Codec.InputMode.OFF},
    new SensorCan12Codec.ActiveWhen[] {
        SensorCan12Codec.ActiveWhen.LOW,
        SensorCan12Codec.ActiveWhen.HIGH,
        SensorCan12Codec.ActiveWhen.HIGH});
```

C++ exposes `ConfigureInputsRuntime` with `std::array<InputMode,3>` and `std::array<ActiveWhen,3>`, and `ReadInputState()` for classification/configuration. `input()` / `ReadInput()` require fresh, healthy status and all inputs settled. Level helpers return UNKNOWN when the channel valid bit is clear. See mode behavior and electrical limits.

8 Purchasing and factory quote

Source: docs/FACTORY_ORDER.md

Use RevE_SC12_Factory_Upload.zip for a **prototype lot**, after reviewing the validation plan. No board has been physically qualified, no factory quotation is included and no order has been placed. The exact-MPN BOM is intended for a Chinese or other assembler that accepts customer-specified sourcing; it is not a pre-approved JLCPCB/LCSC turnkey basket.

For a JLCPCB upload, start with the fabrication-only Gerber ZIP, then supply BOM_JLCPCB.csv and CPL_JLCPCB.csv. Keep Factory_BOM.csv alongside for exact-MPN matching. The larger factory packet is a complete reference handoff.

Fabrication

Setting	Requirement
Board	52 × 70 mm, four copper layers, FR-4, 1.6 mm nominal
Layer order	F.Cu signal/power, In1.Cu GND, In2.Cu GND + three bias-control traces, B.Cu signal/power
Copper	35 µm nominal each layer; verify finished copper with supplier
Dielectric	Symmetric four-layer stack; target outer-to-inner dielectric ≤0.25 mm; supplier must confirm final stack
Finish	ENIG specified for exposed programming contacts and fine-pitch QFN
Minimum trace	0.18 mm
Clearance	0.20 mm general; U8 package pad-to-pad 0.15 mm local minimum
Vias	Through vias, minimum 0.30 mm drill / 0.60 mm pad; no blind/buried vias
Drills	Separate PTH and NPTH Excellon; no plated mounting or alignment holes
Solder mask	Both sides; preserve fine-pitch openings; review U8 mask webs with CAM
Stencil	Top only, nominal 0.10 mm laser-cut; use supplied paste apertures
Panels	Factory rails/fiducials may be added outside the delivered outline; preserve connector overhang

Native nominal stack: 0.035 mm copper on all four layers, 0.20 / 1.04 / 0.20 mm dielectric and 0.01 mm solder mask per side, totaling 1.60 mm. These are planning thicknesses; final fabrication tolerances and material selection belong to the approved supplier stack.

This stack is an order requirement, not a confirmed laminate/impedance quotation. CAN is short board-level routing; no fabricated impedance coupon has been qualified. Get the supplier's CAM/stackup review before fabrication.

Assembly

Factory scope: 111 SMT components on the top side only. Use `bom/Factory_BOM.csv` (exact MPNs) and `output/assembly/CPL_JLCPCB.csv`. The simplified `bom/BOM_JLCPCB.csv` is a matching aid; blank LCSC numbers require exact-MPN matching against Factory_BOM, never an automatic substitution.

User scope: six terminal blocks plus three voltage headers (nine through-hole parts), then three loose caps. The factory must leave J1-J6 and JP101/JP201/JP301 empty. `positions_all_top.csv`, the complete BOM and drawings describe the finished board, not additional factory placements.

CPL coordinates are mm from the PCB lower-left, positive X right and Y up, with native KiCad counterclockwise rotation. Match the uploaded preview to the assembly drawing and check IC/diode pin 1; machine-library rotation conventions can differ. R64 is physically a normal two-terminal 1206 resistor; its four numbered PCB

pads include two copper-only Kelvin taps. NT1 is a copper ground tie and is not a purchased component. Do not interpret pad count as component termination count.

Power headers are unpopulated; the JP2 termination bridge and all J8 contacts are **unbridged/unpasted as supplied**. The factory must not add solder across JP2 unless a separate reviewed port-configuration work instruction is supplied. The user will hand solder all through-hole parts after delivery. The family name “LSF-SMT” does not make these exact connectors ordinary two-pad SMT parts.

Do not replace U8 with a generic 21-pin QFN, substitute a lower-saturation inductor, change the precision divider grade, or replace the 1210 capacitor bank with a nominally equal capacitance without approval of the actual electrical/footprint data. DigiKey substitute labels alone are insufficient. Maintain 0.1 ohm shunt value, tolerance and terminal orientation.

Include inspection of fine-pitch joints and the five long QFN power pads with suitable inspection/X-ray for the prototype lot. Perform unpowered shorts/selector tests, current-limited staged bring-up, firmware programming/readback and functional rail/input/CAN tests before accepting units. Track lot and component substitutions.

Sourcing, cost and delivery

Cost reduction and batch estimates separates parts procurement from assembly. LCSC inventory is not proof of JLCPCB factory allocation. No SMT part allocation, complete cart or final quotation has been confirmed. Request exact-MPN matching and incoming date confirmation before paying; the STM32 and TPS552872-Q1 may require outside procurement. Any unapproved item must remain unresolved rather than being silently substituted.

Request Economic PCBA first if the exact parts and four-layer board qualify; compare Standard PCBA with the same SMT-only scope. Avoid paying for bottom-side assembly or through-hole work. Standard assembly still charges feeder loading for all 46 SMT types. No programming or functional-test service is included by this package unless separately quoted; the user performs bring-up and programming.

A delivery within one month is plausible with allocated parts and tracked air express. Budget 2-4 weeks as a planning allowance, not a carrier promise; procurement, review holds and customs can exceed it. Country/postcode and order quantity are needed for an actual delivered quote. Have the manual parts arrive in parallel. Do not select a long supplier lead time to obtain a lower component price when the one-month target matters.

Programming deliverables

The packet includes combined `firmware/build/factory.hex` and `tools/program_factory.tcl` for initial SWD programming/protection. Power cycle and verify protection, then use CAN for setup and later `application.sc12` updates. User programming is the default quoted scope; see fixture procedure. Do not substitute an application-only ELF/HEX.

9 Validation and qualification

Source: docs/VALIDATION.md

No Rev E prototype has been assembled, powered or tested. The files support engineering review and prototype fabrication. The production/competition qualification gates below remain open, even when every automated check passes.

Automated evidence

The release contains native KiCad electrical/design-rule reports, explicit pin-model/schematic/PCB parity, placement/selector/programming-contact checks, manufacturing-file parsing and copper checks, host firmware tests, and Java/C++/Python protocol tests. These check design consistency and modeled behavior. Mocked peripheral tests do not prove STM32 timing or CAN electrical operation.

Electrical checks sweep input-threshold tolerances, divider accuracy, resistor stress, fuse/load budgets and manufacturer-curve capacitor derating. The separate converter model screens boost CCM loop corners only. Neither substitutes for hardware measurement. Reports state their assumptions rather than treating typical values as guaranteed bounds.

Prototype gates before production release

Gate	Measurement / acceptance intent
Assembly	Confirm exact horizontal terminals, pin-to-hole fit, polarity, U8 long-pad joints, R64 Kelvin layout, selector isolation and no solder shorts
Staged power	No sensors first; current-limited 12 V source, verify 5 V/3.3 V before full operation, then regulated 12 V and low-load input current
Regulation	Sweep 6-22 V with 0-300 mA on each rail separately and together; verify 12 V remains within the selected sensors' supply limits at their cable ends
External bias	Scope V5_INTERNAL, BB_EN, V12_RAW and 3V3 during slow/fast input ramps, repeated starts and brownouts. Confirm converter bias never causes unsafe switching or back-powering; review operation below 6 V
Transients	6↔14↔22 V line steps; 0↔300 mA load steps; representative sensor startup/inrush and cable lengths; check ringing/undershoot through the buck/boost crossing
Stability	Measure loop response or justified equivalent transient qualification in buck, boost, transition and light-load/PFM modes; target >45° phase and >10 dB gain margin where measured
Faults	Short each sensor supply and each port; verify current-limit behavior, 12 V automatic recovery versus 5 V latch, firmware flags, no spurious counts and safe temperatures. Do not assume every 12 V short enters hiccup
Electrical protection	Reverse input, realistic protected-branch fault energy, fuse clearing/I^2t, inductive transients and TVS stress using suitable test equipment. Not automotive load-dump qualified
Temperature	Both rails fully loaded at 6/12/22 V, intended ambient and later enclosure; measure U8/U1/L1/L2/Q1/fuse/terminal and MCU temperatures, hot-start/restart and surface safety
Inputs	0-30 V PNP/NPN/push-pull thresholds, off-state leakage/residual voltage, ground offset, sensor disconnection, wiring transients and all intended sensor models
CAN/firmware	Real roboRIO bus at 1 Mbit/s with correct end termination; no normal telemetry before provisioning, unique IDs, bus-off, clock faults, watchdog, broadcast disable, explicit resume, stale data and power-fault handling
EMC/mechanical	Motor-noise environment, ESD at terminals, cable retention, vibration, mounting insulation and programming-adapter fit; repeat after enclosure design
Production process	Approved stackup/CAM, stencil/profile, supplier alternates, test fixture, serial/lot records, programming/provisioning and final functional limits

FRC integration

Use a single appropriate protected power-distribution branch and wiring sized for its protection. The 0451 3 A fuse has a published 300 A interrupt rating at 32 V DC; confirm prospective branch fault current and clearing energy for the actual harness. The onboard fuse is secondary protection, not permission to bypass the robot distribution rules. Sensor grounds are common with the board; isolate conductive sensor cases where required by frame-ground rules. Do not use dedicated radio/roboRIO power outputs as a convenient supply.

The custom Team Use CAN address uses manufacturer 8 and miscellaneous device type 10. The implementation processes broadcast disable ID 0; it has no actuator outputs. It is not a CTRE Phoenix device. Robot inspection and the current season's rules still apply. Commercial CAN product acceptance, assigned identifiers and required ecosystem support are separate outstanding steps; Java/C++ examples alone do not establish approval.

Primary references: 2026 FIRST manual, especially custom-circuit voltage, distribution/wire and frame-ground rules; WPILib CAN addressing and acceptance requirements. Recheck applicable rules for the actual event/season.

Rev E-specific prototype checks

Verify the exact 2 mm header and cap fit, both 5 V / 12 V positions and unpowered open state on every channel. Verify no connection between the two rails in either position. Check solder fill on all 39 user-fitted through-hole joints. Fit/retention, vibration and impact testing must include the actual caps, chosen removable retention compound, enclosure and robot harness; these headers have no positive latch. Compare the substituted passive values/ratings against the supplied BOM. Factory BOM and CPL must contain exactly 111 SMT references, with all nine through-hole components omitted. Firmware identity is hardware 5 / protocol 4; prior-revision firmware is rejected.

CAN servicing qualification

The service engine and host uploader pass simulated flash cuts/retries; ARM register checks and both bridge adapters compile. Before robot use, measure actual boot/vector/interrupt behavior, option-byte protection, stack use, upload time, CAN load, bus-off recovery and duplicate-ID discovery. Interrupt power during invalidation, every erase/program phase and final commit on physical units. Verify persistent configuration, bootloader recovery, controller disconnect/enable rejection and healthy telemetry after exit. Protocol and evidence; per-unit fixture records. No software simulation closes these physical gates.

Electronic bias qualification

Verify Q101/201/301 PMOS and Q102/202/302 NMOS polarity and correct exact MPNs. Scope all six gate signals during reset, SWD programming, bootloader, watchdog recovery and supply ramps: pull-up gates HIGH and pull-down gates LOW mean both biases off. Verify mixed NPN/PNP/push-pull operation and nonvolatile readback independently on all three channels.

Sweep external signal 0-30 V in every mode. Measure inactive sensor leakage margins, active residual voltage and pull-up sink current (~3.5 mA maximum). Test 3.3 V and 5 V push-pull, unpowered sensors and unpowered board backfeed. Qualify the BAT46W leakage/rail-bleed budget over the actual operating temperature; current calculation uses its 60°C reverse-leakage specification, not a zero-backfeed claim.

Test floating detection with open contacts, NPN and PNP inactive outputs, long cables and added capacitance. Verify UNKNOWN during inconsistent probe pairs, no probe-generated edge counts, and one debounced edge on actual active-state transitions. Check keeper retention while released and LOW initialization after reset. Verify fixed input modes for faster signals; float detection can alias changing sources.

10 Connector evidence

Source: docs/CONNECTORS.md

References	Exact MPN	Series description	DigiKey
J1-J3	**1824420000**	LSF-SMT 3.50/02/90 3.5SN BK TU	281-2020-ND
J4-J6	**1824430000**	LSF-SMT 3.50/03/90 3.5SN BK TU	281-2021-ND

Manufacturer pages: 2-pole, 3-pole. `/90` gives horizontal wire entry parallel to the PCB. `/135` is the angled version and is not used. Phoenix PTSM is not fitted.

These are through-hole/THR parts despite “SMT” in the family name. Pitch is 3.5 mm, with two solder legs per pole and 7 mm row spacing. PCB drills are 1.1 mm. Housing widths are 7.7 mm / 11.2 mm. Manufacturer CAD is 14.3 mm deep; the footprint conservatively reserves 14.75 mm. Height is 8.5 mm above the mounting surface; nominal pin tips reach 3.5 mm below that surface (about 1.9 mm beyond a 1.6 mm PCB).

Current renders use the official STEP solids with a rigid translation (-2.13,-12.625,-3.5) mm into the footprint origin. They are not approximate connector boxes. Display colors are assigned by the project. Original files, transformed files, URLs, hashes, pin/drill checks and nominal bounds are in vendor_connectors.json. Script `import_connector_models.py` checks all six placed pad patterns and the two vendor models. Do not regenerate models merely to change routing: STEP exporters can alter file headers without changing geometry.

Top terminals face outward at the power/CAN edge; three sensor terminals face outward at the opposite edge. Conservative complete footprint envelope is 52 ×78.1 mm; add room for cable bend, strain relief and button access. Renders do not validate conductor retention, tool clearance in an enclosure, molding tolerances or actual solder fit.

Rev E voltage headers

JP101/JP201/JP301 use Harwin M22-2010305: 2 mm pitch, 0.50 mm square pins, 6 × 2 mm body, nominal 5 mm height above the PCB and 3.4 mm tails. The project’s 2 mm header footprint uses 0.8 mm drills and a 0.25 mm nominal body courtyard margin. Pads 1-3 run left-to-right at Y=57.5 mm; pin-1 positions are X=5, 21 and 37 mm.

The Sullins SPN02SVEN-RC drawing specifies a closed top, 2 mm pitch, 0.50 mm square mating pins and nominal 4 × 2 × 4.92 mm body. Contact height is 2.45 mm from the entry, within the header’s 3.5 mm mating pin length. Nominal mating geometry agrees; physical fit and retention remain untested. Caps are delivered loose and not included in the assembled-board STEP model. Use the finished cap envelope for enclosure access/clearance and allow room to remove them.

The generic local header STEP represents the nominal pin/body geometry, not manufacturer-qualified tolerances. Vendor connector hashes continue to cover the unchanged six terminal models. See wiring for one-cap-per-header configuration and vibration retention qualification.

11 Tools and reproduction

Source: TOOLS.md

The saved native KiCad schematic/PCB are authoritative. Generation/placement/router scripts record design construction; they are not a one-command reroute of the reviewed board. Editing the circuit requires matching schematic, BOM, PCB and new checks. Earlier revisions remain separate.

Tested toolchain: KiCad 10.0.6, its matching Python/pcbnew interpreter, Python 3.12, Arm GNU GCC 13.2.1, GNU make, host GCC/G++, Java 17, Poppler, InteractiveHtmlBom 2.11.2 and the packages pinned in `tools-requirements.lock`. Tool binaries/caches are external to the repository. Install those native tools for the host operating system. The `kicad-python` wrapper must use the interpreter matching the KiCad binary bindings, not an arbitrary venv Python.

```
# In rev_e/, after installing native tools:
uv venv --python python3.12 .venv
uv pip sync --python .venv/bin/python tools-requirements.lock
source tools/env.sh
python tools/fetch_driver_dependencies.py
bash scripts/validate_release.sh
```

`tools/env.sh` uses a revision-local venv, or the repository venv when working in the full checkout. It prepends `$CANDI_TOOLS_ROOT/bin` (default `$HOME/.local/share/candi-tools/bin`). Wrappers expected there are `kicad-cli`, `kicad-python`, `host-gcc`, `host-g++`, `ibom`, and Arm compiler commands. Java is under the configured tools root or selected with `CANDI_JDK`. Fonts for the PDFs are DejaVu Sans/Mono.

The validation script checks the saved design and refreshes manufacturing evidence, firmware and host tests; it does not move or reroute components. A visual/documentation refresh is separate:

```
bash scripts/render_board.sh
python scripts/build_documents.py
python scripts/build_handbook.py
python scripts/review_documents.py
python scripts/build_release.py
```

The factory packet is built by `build_documents.py`. Rebuild it after any BOM/placement/fabrication change. `build_release.py` selects authoritative files and writes a manifest plus ZIP checksum. Check an extracted archive with `verify_archive.py`; the release manifest deliberately excludes itself and the enclosing ZIP from its file hashes.

Use `python scripts/import_connector_models.py` only when restoring/reviewing vendor models. It checks the original STEP hashes and all six native connector pad patterns, then writes translated local models. It can change STEP headers, requiring a fresh STEP export, renders and release. Keep the originals and provenance JSON together.

`hardware/design/circuit.py` is the explicit component/net model; `generate_schematic.py` rebuilds the schematic. Construction helpers such as `pack_compact.py`, `route_critical.py` and `apply_routes.py` operate on intermediate files and can overwrite placement/routing. They are not part of release validation. The boost CCM analytical loop screen is in `check_converter_loop.py`. It is not a switching simulation or thermal qualification. Do not reuse older revision reports as evidence for this changed supply.

In the full repository, `kicad-python scripts/check_variant.py` compares the electronic-bias circuit delta and critical routing with the preserved neighboring `rev_d2/`. The dated result is included in variant_verification.json; a standalone E archive does not contain the older D2 baseline.

For CAN servicing also run `CANDI_JDK25=/path/to/jdk25 bash scripts/test_bridge2027.sh`. Dependencies are pinned in `tools/bridge2027_dependencies.json`; Java 25 is required by 2027 alpha-7. `test_firmware.sh` includes C upload-engine fault injection and compiled ARM register tests. `compare_assembly_cost.py` generates partial-fee scenarios, not live quotes.

12 Sources and notices

Source: docs/SOURCES_AND_LICENSES.md

KiCad library/model notices are retained under `licenses/`, together with Arm CMSIS, ST device-header and WPILib notices. Firmware third-party directories and InteractiveHtmlBom output retain their notices. Original project work has no selected blanket license. Manufacturer documents/CAD remain their owners' material; retaining a file is not a redistribution-license grant.

Subject	Primary reference
Input fuse interruption / derating	Littelfuse 451/453
Buck and capacitors	TI LMR51420
Regulated 12 V converter / exact RYQ0021B package	TI TPS552872-Q1, SLVSHF9 August 2023
Shielded 3.3 µH inductor	Bourns SRP5030TA
Input/output/bias capacitor curve	Samsung CL32B106KBJNNW
Banner supply/residual voltage exceptions	QS18 product manual
5 V latch-off limiter	TI TPS2553, -1 variant
Comparator/leakage caveats	TI LM339LV
CAN interface / bypass	TI TCAN3413
STM32 pinout, supplies, clock	ST STM32F042
Example industrial output residual levels	Omron E3Z specifications
Terminal geometry and exact MPNs	Weidmüller 1824420000, 1824430000
CAN addressing / broadcast disable	WPILib

Samsung CL31A226KAHNNNE typical characteristic data is retained in `sources/datasheets/samsung_CL31A226_characteristics.json`, inherited from the earlier design's manufacturer retrieval. Typical curves do not establish guaranteed capacitance. DigiKey URLs, exact codes and dated observations are in `bom/catalog_selected.json`. Vendor STEP URLs, hashes and transforms are in `hardware/library/3d/vendor_connectors.json`.

Equivalent parts need engineering review; distributor substitution suggestions do not establish electrical/mechanical equivalence. Availability and season rules can change after these observations.

Rev E retains the retrieved Samsung CL32B106 characteristic JSON and TI converter datasheet under `sources/datasheets/`. The project-created converter footprint/model is documented in `hardware/library/converter_package.json`; its STEP representation is parametric package geometry, not official manufacturer CAD. The horizontal connector STEP models are official vendor geometry.

Rev E sourcing URLs and tiers are in catalog_china.json. Header/cap geometry links the Harwin and Sullins manufacturer drawings; cost assumptions links the assembly fee and lead-time sources. Stock observations are not factory allocations.

Rev E input bias references: AO3401A, BSS138BK, BAT46W. The local SOT-23 footprint/model comes from installed KiCad libraries under their existing library-license policy. CTRE input-mode API informed feature choices; no CTRE firmware or hardware design is included.

13 Cost reduction

Source: docs/COST_REDUCTION.md

Rev E preserves D2's lower-cost passive sourcing, top-side SMT assembly and user-soldered through-hole parts. It adds electronic bias: six MOSFETs, six gate resistors and one rail-bleed resistor. Three existing pull-up resistors change to 1 kΩ, using an existing BOM MPN. The three unpurchased bias solder selectors disappear.

Boards	Total parts purchase per board	Difference from D2
1	$56.42	+$1.15
5	$40.96	+$0.56
10	$35.14	+$0.45
25	$32.72	+$0.49

These September 8, 2026 estimates include supplier minimums, terminals, power headers and three loose caps. PCB, assembly, attrition, programming/test, freight, tax and duty are extra. First-tier consumed parts are $44.69; purchasing enough minimum packs for one board costs $56.42. Batch purchasing remains the strongest saving without changing the product's hardware functions.

Cost CSV, supplier observations, BOM and assembly scenarios expose the assumptions. AO3401A is LCSC C15127; BSS138BK,215 is C282529. The six MOSFETs consume about $0.49 at first supplier tiers. Stock observations are not factory allocations; these exact parts still need confirmed sourcing.

The board grows from 52 × 64 to 52 × 70 mm, a 9.4% area increase. All 111 SMT parts remain on top, with nine through-hole parts hand soldered after factory assembly. There are 48 SMT MPN types, two more than D2. This adds $3.06 per order in the published Standard feeder-fee model, plus placement and any quote adjustments. No second-side assembly fee is introduced. The published assembly fee schedule is an estimate input, not a quotation.

The main remaining expenses are the exact MCU, regulated 12 V power stage, horizontal connectors, multilayer PCB and small-order assembly overhead. The comparator is retained because it gives the existing digital threshold/hysteresis behavior across the input range. Removing it or the power protection solely to lower the BOM would require a separate electrical redesign and qualification.

Request the exact-MPN SMT quote, confirm Economic/Standard eligibility, component allocation and destination-specific express shipping. China delivery within one month is not confirmed; the earlier 2-4 week allowance is a planning estimate subject to sourcing, production and customs. No order or supplier contact has been made.

14 Optimization progress and qualification

Source: docs/OPTIMIZATION_PLAN.md

Updated 2026-09-08: Rev E adds electronic per-channel bias and protocol 4 on top of the D2 CAN servicing and cost reductions. The PCB, firmware, drivers, setup tool and manufacturing packet change together. Physical qualification remains open; older revisions are preserved.

What was implemented

Work	Current evidence	Remaining gate
Separate bootloader/application	Fits existing 32 KiB STM32; protected 8 KiB boot region, 21 KiB app, manifest, redundant configuration	Real option-byte programming, handoff/interrupts and stack high-water
Discover/blink/persistent settings	Full 96-bit UID selection, readback, duplicate-ID simulation, explicit save/reset	Several physical blank and duplicate-ID boards; power-cycle verification
CAN update/recovery	Actual C engine tested through uploader; 2,148 simulated erase/program cut points, retries, bounds/CRC checks	Physical power cuts, bus-off and actual CAN timing; no rollback
Laptop and controller workflow	Python CLI and Tk GUI; real TCP fragmentation/revocation test with mock CAN; 2026 and 2027 alpha-7 adapters compile	End-to-end roboRIO/Systemcore and desktop adapter testing
Programming/test process	Combined factory image, protected programming script, per-UID acceptance-record tool and measurement template	Build fixture; measure pilot units and programming/test labor
Cost	Exact existing MPNs retained; purchase-quantity optimizer and assembly-fee sensitivity table generated	Allocated exact-MPN quotes, quantity/destination, lead time and full delivered price

Use CAN service for protocol and operating instructions, fixture for unit records and cost reduction for current numbers. Compilation and simulations do not close the physical gates. No PCB redesign or new bill-of-material component was needed for servicing.

Lessons taken from LaserCAN and GrappleHook

LaserCAN separates its application and CAN bootloader while retaining SWD for initial installation/recovery. Its STM32F103 bootloader reserves 20 KiB and uses different vector handling; SC12 implements its own compact Cortex-M0 C engine and SRAM remap. Sources: pinned bootloader, memory map, SWD instructions and ST RM0091.

GrappleHook demonstrates serial-based identification/settings, acknowledged offset uploads and a TCP-to-CAN robot path. SC12 provides those architectural capabilities with its own protocol and UID discovery. Our bridge remains inside running robot code, requires opt-in plus disabled state, and binds localhost for SSH access. Reference: pinned management/upload code and roboRIO provider. No reference implementation code was copied; both repositories carry GPL-3.0 notices. GrappleHook does not recognize SC12.

SC12 continues to serve three external industrial digital inputs. LaserCAN's integrated ranging sensor has different power/interface needs, so its board size and price are not like-for-like targets. SC12 retains regulated 5 V/12 V at 300 mA shared per rail, 6-22 V input target, top-side-only assembly, removable power caps and no slide switches.

Next qualification loop

1. Program two pilot boards through SWD; verify protected pages and configuration retention. Measure boot/startup, watchdog recovery, vector interrupts, stack headroom and update time. Confirm the firmware truly returns to healthy telemetry after maintenance.

2. Test duplicate IDs, blank devices and mixed boot/application states on a real CAN network. Disconnect controller/host, interrupt power at erase/program/commit, corrupt images and force bus-off. Repeat uploads and verify configuration and bootloader integrity. Verify the controller bridge's permission loss and robot output interlock.

3. Exercise representative Banner/Omron models at cable ends: both rails independently/together to 300 mA, 6/12/22 V input, startup, shorts, brownout and enclosure temperature. Validate sensor residual-voltage limits rather than assuming all NPN sensors work.

4. Qualify terminal, mounting, harness and actual cap/retention compound under vibration/impact. An absent cap or disconnected sensor can resemble a valid inactive input; do not claim disconnect detection. Consider a passive CAN bypass harness for board replacement without opening the robot backbone.

5. Record all observations per UID and lot, correct failures, repeat affected tests and publish the measured limits. Complete the validation matrix before production use.

Cost decisions

Keep the reviewed circuit for the pilot. The software fits the current MCU without external memory or a USB connector. The small implemented purchasing correction saves $0.48 across five boards by buying cheaper ten-packs for three precision resistor MPNs; it does not materially change the ten-board price. Parts remain $34.69/board at ten and $32.23 at 25, before assembly/PCB/shipping.

The useful next step is an exact-MPN quote comparison at 5/10/25 boards. Standard fees charge all 46 SMT types; Economic charges extended types differently. The scenario CSV keeps unknown fees separate, with no assumed free feeders. At 25 boards Standard's parts-plus-setup/feeder/fixture subtotal is about $37.05/board versus $46.75 at ten, still excluding fabrication, placement, inspection, freight and other charges. Buying more boards lowers unit allocation while increasing cash for the order.

Connector redesign remains a candidate, not a substitution: compare complete latched-crimp harness cost including housings, contacts, crimp tools/labor, spare harnesses and strain relief. Verify mating cycles, wire gauge, current, polarity, footprints and vibration. Current terminals cost about $7.85/board at ten; the cheaper exact-MPN LCSC listing reviewed had zero stock and is not a basis for promised savings or delivery.

A converter redesign must compare the complete power circuit and requalify regulation through battery sag, thermal behavior and protection. Keep four copper layers and the critical switching/return layout until an actual quote demonstrates worthwhile savings. Preserve effective bulk/local capacitance and precision compensation values. A larger MCU or CAN-FD controller requires a separately justified feature and new hardware revision.

China delivery inside a month remains conditional on allocated components, review/manufacturing time, destination-specific air shipping and customs. No order has been placed, no supplier contacted and no delivered price/date confirmed by this work.

15 CAN setup and updates

Source: docs/CAN_SERVICE.md

Rev E firmware **0.6.0**, bootloader **1.0.0**, service protocol **1**. Normal sensor messages remain protocol **4**, hardware **5**. Implemented and software-tested; actual STM32/robot-controller, power-cut and vibration qualification is pending. This is our own service protocol, not GrappleHook-compatible firmware.

First installation and recovery

Install factory.hex once through SWD using program_factory.tcl. This combines the protected bootloader, relocated application and image manifest. Rev E requires its own factory image and fresh configuration. D/D2 hardware cannot gain the new bias circuit through firmware. The script preserves the two configuration pages and refuses an unexpected device/flash size or an already protected bootloader. Its hardware execution is a pilot qualification gate.

From `rev_e`, with J8 wired per the custom mapping and a separately powered, current-limited board:

```
openocd -f interface/stlink.cfg -f target/stm32f0x.cfg \
  -f tools/program_factory.tcl
```

Power cycle afterward and verify `boot_protected: true` using the tool. Write protection covers flash pages 0-7 (OpenOCD protection blocks 0-1). Updating requires those blocks locked and application/configuration blocks writable. The script does not change readout protection or mass-erase configuration. Do not program `sensor_can.elf` or `.hex` alone as an initial install: these are relocated application-only files without the boot-valid manifest. Do not upload `factory.hex` over CAN.

Normal CAN updates use application.sc12. If interrupted, retry the complete file: there is no old-image rollback or cross-session resume. Invalid images remain in the bootloader. A normal boot gives a 500 ms discovery opportunity; an explicit boot request stays there, and three successive watchdog resets before two healthy application seconds also request recovery. `run` exits a valid bootloader image. SWD remains the recovery path for bootloader/option-byte damage or a nonfunctional CAN connection.

Laptop and robot use

Use Python 3.10+; the robot tunnel transport needs only the standard library. SocketCAN additionally needs `python-can`; configure the adapter for classical CAN at 1 Mbit/s. Tk is required only for the desktop window (`python3-tk` on many Linux distributions).

The robot bridge runs in a background thread alongside robot code. Copy `ServiceBridge.java` and the adapter for your WPILib year from examples/bridge. It listens only on `127.0.0.1:5810`; access it over an authenticated SSH tunnel. The 2026 adapter requires an explicit maintenance-request flag **and** Driver Station disabled state for every transmission. The 2027 alpha-7 adapter uses `RobotState` and an explicit `CANPort`, compiled separately with JDK 25. Choose a classical-CAN-compatible bus; this MCU cannot transmit CAN FD.

Keep the robot disabled, mechanically secure it and keep outputs inhibited in your robot code throughout maintenance. The bridge rejects or disconnects when permission/disabled state is lost; it cannot prevent someone enabling the robot at the Driver Station. No robot program is stopped or redeployed by this tool. Do not expose this unauthenticated TCP framing directly on a network. The sensor protocol itself has no authentication; CRC detects corruption, not a malicious uploader.

Example using a roboRIO reached over USB; use your controller's actual address/account:

```
ssh -N -L 5810:127.0.0.1:5810 lvuser@172.22.11.2
python host/can_service.py --bridge localhost:5810 discover
python host/service_gui.py
```

Copy the desired 24-hex-digit UID from discovery. Replace `UID` below with it:

```
python host/can_service.py --bridge localhost:5810 blink --uid UID
python host/can_service.py --bridge localhost:5810 configure \
  --uid UID --device 17 --invert 0 --debounce 2 --modes npn,pnp,push-pull
python host/can_service.py --bridge localhost:5810 update \
  --uid UID --file firmware/build/application.sc12
```

Direct SocketCAN uses `--channel can0`. Other commands: `info`, `run`, `rate --milliseconds 10|20|50`. Settings persist after save/reset; status period resets to 10 ms. Existing driver timeout is 100 ms. Traffic is 231/131/71

frames/s at 10/20/50 ms; verify bus jitter physically.

Discovery resolves the complete STM32 96-bit UID independently of assigned CAN IDs, including blank and duplicate-ID devices. Blink confirms physical selection. Before a change, the tool reads back all three UID words. ID conflict checks cover **responding SC12 devices only**; a powered-off board or another Team Use product can still have a conflicting address. Use one maintenance client per CAN network. The service APIs must be reserved against other Team Use devices.

Opening a session suppresses normal telemetry on all participating SC12 boards, and freezes/rebaselines input counts. Sensors remain powered by hardware. Drivers may retain their last sample until its normal freshness timeout; robot code must treat maintenance as unavailable immediately. Close/lease expiry restores normal sampling/telemetry unless Broadcast Disable previously latched it off. Broadcast Disable cancels maintenance and pending transmissions; explicit resume remains necessary for that latch. Never interpret a missing sample as a false input.

Wire protocol

Classical CAN, extended identifiers, eight data bytes, no RTR. Request ID `0x0A08C000`, response ID `0x0A08C040` (Team Use 8, miscellaneous type 10, APIs 0x300/0x301, address 0). Payload: opcode byte, sequence byte, nonzero little-endian session token (16 bits), little-endian value (32 bits). Response opcode OR 0x80 means success; OR 0xC0 means error. Echo sequence/token and validate the expected value. Errors 1-5 mean argument, state/offset, flash, image, and missing boot protection. No reply means an unmatched session/selection or transport failure.

Operation	Value / result
OPEN 0x01	Magic 0x314D4353; broadcast ACKs are identical; five-second lease
PREFIX0/1/2 0x02-04	Three raw UID words; no ACK; each change clears selection
PROBE 0x05	Prefix length 0-96, MSB first within each UID byte; matching nodes reply 1; only 96 selects
CLOSE 0x06	End session, no ACK
INFO 0x10	Index 0 flags/versions, 1 settings, 2 mode version, 3 upload offset, 4 capacity, 5 image validity, 6-8 UID words, 9 persistent input configuration
BLINK 0x11	Zero; three-second identify LED
SAVE 0x12	ID/invert/debounce/reserved bytes; reserved must be zero; explicit save then reset
BOOT / RUN 0x13/14	Zero; enter bootloader / run verified application
RATE 0x15	10, 20 or 50 ms; volatile status period
INPUTS 0x16	Stage validated per-input configuration; commit with SAVE; cleared on new selection/session
SIZE / CRC 0x20/21	Padded application length and CRC32; bootloader only
BEGIN 0x22	Product/layout tag 0x53433135; invalidate/read back manifest before application writes
OFFSET / DATA 0x23/24	Byte offset then four image bytes; ACK next offset after flash readback
FINISH 0x25	Zero; validate complete flash CRC/vectors, commit manifest last, ACK image CRC

INFO index 0 packs service version in bits 0-7, bootloader mode in bit 8, configured in bit 9, boot protection in bit 10, application minor 6 in bits 16-23, and hardware 5 in bits 24-31. Index 1 packs ID/invert/debounce/status-period as bytes; unconfigured ID is 255. Index 2 is major/minor/patch in bits 16/8/0. Index 5 returns 0 or 1.

Host discovery walks binary UID prefixes; matching responders send identical presence frames. Prefix frames are paced; UID readback guards selection before writes. Queries retry four times, probes twice; default pacing is at least 6 ms between requests and a 300 ms ACK timeout. Binary discovery can take seconds per device; intermittent connectivity can omit a node, so repeat discovery before commissioning. Uploads send OFFSET and DATA for each four-byte word, bounding throughput and working around flash stalls. Only the

immediately previous exact request is cached; offset/data logic also accepts an identical last word without rewriting it. Gaps, changed duplicate words and stale sessions are rejected.

Image and memory layout

Region	Address / size
Write-protected bootloader	0x08000000 / 8 KiB
Application	0x08002000 / 21 KiB
Image manifest	0x08007400 / 1 KiB
Configuration A/B	0x08007800 / 0x08007C00, 1 KiB each
Reserved SRAM	First 256 bytes: 192-byte vector table, validated recovery cookie

Both linkers reserve at least 1 KiB stack, excluding static data; runtime stack high-water measurement remains pending. The Cortex-M0 application copies vectors to SRAM and remaps them; bootloader handoff clears pending interrupts, stops SysTick/CAN and returns clocks to HSI. Compiled-register tests check vector copying, GPIO/CAN and configuration writes against peripheral models; they do not emulate analog effects or real interrupt timing.

The update file uses `<8s6I`: `SC12UPD1`, format 1, hardware 5, application address, padded length, CRC32, packed version, then image bytes. Length is 192-21504 bytes and a multiple of four. Initial SP must be aligned and within application SRAM, and Thumb reset PC inside application executable bounds. Manifest words are magic 0x49324353, size, CRC, product tag, CRC of the first four words, and final commit 0xC035A19E. Boot verifies manifest, vectors and full application CRC. No update command can write bootloader, configuration or option-byte regions.

Evidence and remaining gates

Tests execute the actual portable C engine with simulated flash, including 2,148 halfword erase/program power-cut positions on a two-page image, retries, corrupt files, wrong sizes, gaps, stale sessions, duplicate IDs and running-version verification. Compiled ARM tests exercise register intent and 34 configuration halfword interruption cases. Real TCP tests cover fragmentation and permission revocation with mock CAN. The Python uploader/GUI, bridge and firmware need end-to-end testing on real hardware before robot use. See qualification and unit programming/test fixture.

The `--modes` option and GUI persist independent input modes and active conditions using INFO 9 / INPUTS 0x16. Omitted modes preserve the existing persistent word; a runtime 0x222 change is not implicitly saved. SAVE commits ID, inversion, debounce and staged input settings together. See input modes.

16 Programming and unit fixture

Source: docs/FIXTURE.md

Build a reusable fixture around the documented custom TC2050 contacts, ST-Link, a separately current-limited J1 supply, accessible rail test contacts, CAN adapter and three independently driven input fixtures. This document and scripts are ready for the pilot; no physical fixture or unit test has been performed.

Leave voltage caps absent for initial programming. Keep the SWD probe's voltage pin as target reference only. Program/verify the combined image and set boot write protection with program_factory.tcl, then power cycle and verify protection over CAN. Save the programmer log, factory image SHA-256, component lot and full MCU UID. The setup checks STM32F04x device ID 0x445, 32 KiB flash and WRP pages 0-7 (OpenOCD protection blocks 0-1, not 0-7), consistent with the OpenOCD STM32 driver. Qualify the exact probe/OpenOCD version on the first unit.

Configure a unique ID through the service tool. Record internal 5 V, logic 3.3 V and regulated 12 V with a calibrated meter; acceptance windows are 4.8-5.3, 3.15-3.45 and 11.4-12.6 V. These generous unit-screening windows do not replace rail regulation/load/temperature qualification. Power off before fitting selector caps. Exercise each input low and high using the approved bias and compatible test signal; observe fresh protocol-v4 status and count transitions. Verify telemetry after closing maintenance. Power cycle again and verify saved ID/inversion/debounce and all three input modes/active conditions.

Copy fixture_measurements.template.json, enter measured voltages and set each check true only after actually observing it. Record live identity/protection plus those results:

```
python host/fixture_record.py --channel can0 --uid UID \
  --measurements measured-unit.json --operator NAME --lot LOT \
  --output unit-UID.json
```

Use `--bridge localhost:5810` for a robot tunnel. An incomplete/failed record exits nonzero. Existing output files are never overwritten. The JSON stores operator/lot/time/UID/settings and a hash of the supplied measurements; keep it with the programming log. It does not independently measure voltages or validate operator assertions.

Design qualification still needs 6/12/22 V input tests, both rails at load, shorts, temperatures, representative sensors/cables, vibration, actual flash power cuts, bus-off/controller disconnect and boot/interrupt timing. See VALIDATION.md. Complete the pilot before calling units production-ready or quoting a qualified manufacturing yield.

Exercise mixed software modes and verify floating detection without false counts using INPUT_MODES.md. Unsettled inputs must not appear as valid robot measurements.

17 Software input modes

Source: docs/INPUT_MODES.md

Rev E removes JP103/JP203/JP303. Each input has its own electronically controlled pull-up and pull-down. Select the mode in the desktop setup tool, CAN service CLI, or Java/C++ runtime API. This is an independent design with CANdi-like input choices; CTRE Phoenix and GrappleHook are not compatible protocols.

Mode in setup tool	Electrical behavior	Default active condition	Typical use
`npn`	Pull toward 3.3 V through a diode and 1 kΩ	LOW	NPN / open collector sinking output
`pnp`	Pull toward ground through 10 kΩ	HIGH	PNP sourcing output
`push-pull`	Both bias switches off	HIGH	Actively driven 3.3 V, 5 V or industrial logic
`float`	Alternate pull-up/down to classify the source	DRIVEN	Slow contact / high-impedance diagnostics
`keeper`	Bias toward the last sampled level; starts LOW	HIGH	Hold a released line near its last logic level

Active condition is independently selectable as `high`, `low`, `driven`, `floating`, `not-high`, or `not-low`. Inversion is applied afterward, then debounce. Normally leave the inversion mask at zero when using these active conditions. Firmware defaults to push-pull/HIGH on all channels; it does not guess a sensor type.

Wiring and electrical limits

Power, ground and signal remain the three sensor-terminal contacts. Choose 5 V or 12 V sensor power with the separate removable power cap, with board power disconnected. These power caps and CAN termination are not input-bias settings. Keep a common signal ground.

The pull-up uses the internal **3.3 V rail regardless of sensor supply**. Its series diode blocks most current from a 5-30 V signal toward the MCU rail. The voltage-rated pull-down MOSFET switches a 10 kΩ resistor. The LM339LV comparator, protected divider and hysteresis still receive the field signal; no field-voltage connection goes directly to an STM32 GPIO. This remains digital sensing, not analog acquisition.

Targets: positive 0-30 V input, LOW ≤1.2 V, HIGH ≥2.4 V, ±0.1 V ground-offset allowance. The compatibility budget allows 100 µA OFF leakage for NPN or PNP sensors. NPN LOW residual must remain ≤1.2 V at up to approximately 3.5 mA sink current. Verify the exact sensor datasheet. A nominal 5 V push-pull signal works with bias off. A 3.3 V open-drain output may use NPN mode if it tolerates the board's pull-up while the sensor is unpowered; otherwise use an external pull-up to its own supply with onboard bias off.

Both switches default off through external gate resistors during reset and in the bootloader. Firmware applies saved modes during startup and suppresses measurements until input/rail settling completes. Removing a power cap does not disconnect SIGNAL; an enabled pull-up can inject current into an unpowered sensor. Set push-pull/bias-off before servicing an externally connected unpowered device. Reverse diode leakage is finite: the 10 kΩ 3.3 V rail bleed budgets up to about 0.61 V unpowered rail rise using the BAT46W 60°C reverse-leakage specification. This is not isolation or a guarantee at higher temperatures.

Floating detection and counts

Float mode waits at least 2 ms after each bias change. Two matching full pull-up/down pairs are required, so initial classification needs approximately 8 ms, plus configured debounce for subsequent logical changes. Input-level telemetry reports HIGH, LOW, FLOATING or UNKNOWN (unsettled). Drivers mark logical inputs unavailable while any channel is unsettled or either rail is unhealthy.

Probe transitions never enter the logical edge counter. Counters track debounced inactive-to-active transitions. They pause during rail faults and maintenance; startup/reconfiguration re-baselines without inventing an edge. Use float mode for slow signals, and qualify the real cable capacitance, noise and sensor leakage. Fast changes during a probe pair can be misclassified; fixed NPN/PNP/push-pull modes are preferable

for ordinary sensing. This is not an encoder interface.

High impedance can mean a disconnected cable, an open contact, or the normal inactive state of an NPN/PNP output. Float detection cannot distinguish these causes. Keeper mode biases a released wire; it does not retain a value across reset or report a disconnection.

Save an example configuration

From `rev_e/`, after discovery and physical identification:

```
python host/can_service.py --bridge localhost:5810 configure \
  --uid UID --device 17 --invert 0 --debounce 2 \
  --modes npn,pnp,push-pull
```

Replace `UID` with the discovered 24-digit hexadecimal serial. The tool stages all three settings, commits with the ID/settings record, resets, and verifies persistent readback. If `--modes` is omitted, existing saved input settings are preserved. Optional `--active low,high,high` overrides the defaults. The desktop window offers the same choices independently for each channel.

Java `configureInputsRuntime(InputMode[], ActiveWhen[])` and C++ `ConfigureInputsRuntime(...)` apply volatile settings. Read back `inputState()` / `ReadInputState()` to verify the configuration word and classification; runtime CAN commands have no acknowledgment and do not write flash. Use CAN service to persist. See wire format, service and drivers.

Engineering checks and simulations do not qualify electrical behavior. Bench-test all modes across supply, temperature, cable length, hot-plug, sensor startup and robot vibration before deployment. See validation.